

Bayesian Interpretation of BLAST Evidence

Zane Perry

1 May 2026

1 Introduction

BLAST, the Basic Local Alignment Search Tool, is one of the standard tools in computational biology for finding local sequence similarity in DNA and protein databases. Its appeal is both biological and computational. Biologically, similar sequences can indicate shared function, shared structure, or shared evolutionary origin. Computationally, BLAST provides a fast heuristic alternative to exhaustive alignment while still returning scores and database-calibrated statistics that are highly useful in practice. The original BLAST framework and its later gapped extensions remain foundational references for modern sequence search. [1, 2]

This project started from a simple statistical question: what does an E-value actually tell us, and what does it not tell us? In ordinary BLAST output, the E-value measures the expected number of chance matches at least this good under a null model of random sequence similarity. That is useful, but it is not itself a posterior probability that a sequence is homologous. The main goal of this project is to build a Bayesian reinterpretation of BLAST evidence that converts E-value information into posterior summaries on a parameter scale that is easier to interpret probabilistically. The point is not to replace BLAST’s scoring theory, but to place its output inside a Bayesian model whose posterior can be explored numerically.

The project uses two connected case studies. The first is a protein-level analysis centered on hemoglobin subunit beta. Several query proteins are compared against a fixed human hemoglobin beta target, including synthetic mutation ladders and a real chicken homolog. The second is a DNA-level sequential updating experiment based on the human beta-globin locus. There the genomic sequence is split into short fragments, each fragment is searched against species-specific databases, and the posterior is updated as evidence accumulates. These two examples emphasize different strengths of the Bayesian framework: the protein example is a controlled fixed-target comparison, while the DNA example highlights sequential updating over multiple fragments.

The data used here are observational rather than experimental. Protein sequences were taken from UniProt, while the genomic locus was drawn from NCBI RefSeqGene resources. The resulting report therefore focuses less on experimental randomness in data collection and more on statistical modeling of search evidence derived from curated biological databases. UniProt and RefSeq are appropriate sources for this because both are standard curated resources used widely in biological sequence analysis. [3, 4]

The main questions of interest are:

1. How should BLAST evidence be transformed so that it can be modeled with a simple Bayesian likelihood?
2. How does posterior support change as a protein query becomes more heavily mutated relative to a fixed hemoglobin target?
3. Can the same framework be extended to sequential DNA fragment evidence for species-level gene matching?
4. What limitations appear when this Bayesian reinterpretation is used as a first proof-of-concept rather than a full generative homology model?

2 Background on BLAST and Statistical Motivation

BLAST performs local rather than global sequence matching. This matters because two sequences may share only one conserved region while differing substantially elsewhere. For protein searches this is especially important, since functionally meaningful similarity is often concentrated in motifs or domains rather than across an entire sequence. BLAST supports several program types, including `blastn` for nucleotide–nucleotide comparison, `blastp` for protein–protein comparison, `blastx` for translated nucleotide queries against protein databases, and `tblastn` for protein queries against translated nucleotide databases. [1, 2]

BLAST scoring reflects biological rather than purely literal similarity. In DNA alignment, a simple model rewards matches and penalizes mismatches and gaps. In protein alignment, substitutions are not scored uniformly, because some amino-acid replacements are more conservative than others. This is why protein BLAST uses substitution matrices such as BLOSUM62, with PAM and BLOSUM representing the two best-known historical matrix families. Gap penalties are affine, meaning that opening a gap is penalized more heavily than extending an existing one. [5, 2]

If the raw alignment score is denoted by S , then BLAST converts that score into a normalized bit score using the Karlin–Altschul calibration constants K and λ :

$$S' = \frac{\lambda S - \ln K}{\ln 2}.$$

The corresponding E-value can be written in raw-score or bit-score form as

$$E \approx K m n e^{-\lambda S} \quad \text{or} \quad E \approx m n 2^{-S'},$$

where m and n denote effective query and database lengths. These formulas explain why bit scores are convenient for comparison across searches and why E-values inherit an explicit null-model interpretation. [1, 2]

The key statistical motivation for the present project is that an E-value is not a posterior probability of homology. A very small E-value indicates that the observed match would be surprising under the null model, but it does not directly report the probability that the hit is a true homolog. That gap between null-model significance and posterior interpretation is exactly where the Bayesian model enters. Rather than asking only whether the observed match is surprising under randomness, the Bayesian approach asks how strongly the observed BLAST evidence should update prior uncertainty about a parameter representing homolog-like evidence.

3 Data and Study Design

The report uses two related datasets.

The first dataset is protein-based and centers on human hemoglobin subunit beta. Human beta-globin is a 147-amino-acid component of hemoglobin, the oxygen-transport protein in red blood cells. The fixed target sequence is the UniProt human hemoglobin beta accession P68871. A real chicken beta-globin sequence, accession P02112, was also included as a biologically meaningful nonhuman homolog. In addition to these two real proteins, four synthetic mutation ladders were created from the human target by randomly substituting amino acids at rates of 10%, 25%, 50%, and 75%. All query proteins were then compared against the original human beta-globin sequence as the fixed target. This creates a controlled spectrum from exact identity, through progressively degraded synthetic variants, to a real homolog from another species.

The second dataset is DNA-based and centers on the human beta-globin locus. The source sequence is the human *HBB* RefSeqGene record LRG_1232 on chromosome 11, with total length 10106 base pairs. Because BLAST is less useful when the entire long locus is treated as one monolithic query for the purposes of this project, the sequence was split into five shorter non-overlapping fragments of length 200 bp. Each fragment was then searched with `blastn` against species-filtered databases corresponding to *Homo sapiens*, *Pan troglodytes*, and *Mus musculus*. The top hit for each fragment/species combination supplied the E-value used in the Bayesian update.

The biological and statistical logic of the two studies is slightly different. In the protein case, the question is how posterior support changes as a fixed target is queried by increasingly dissimilar proteins. In the DNA case, the question is how posterior support for species membership accumulates when multiple fragments from one genomic locus are processed sequentially. Both settings are observational, because the sequences are drawn from curated public databases rather than generated by a designed experiment. The sequential DNA study is especially useful because it turns one long genomic sequence into a series of smaller evidence contributions, which makes Bayesian updating conceptually natural.

4 Bayesian Model and Computation

The first step is to transform BLAST E-values into a scale that is easier to model continuously. Let

$$E^* = \max(E, 10^{-300}), \quad Y = -\log_{10}(E^*).$$

This transformation handles BLAST-reported zeroes by flooring them at 10^{-300} and then mapping all E-values to a real-valued evidence scale. On this scale, smaller E-values correspond to larger Y . For example, an E-value of 10^{-6}

becomes $Y = 6$, while an E-value near 1 becomes $Y = 0$. This transformation was chosen because it is interpretable and allows a normal likelihood to be used naturally. The flooring step also ensures that extremely small reported values remain numerically tractable.

The prior model places a Beta distribution on a parameter $\pi \in [0, 1]$:

$$\pi \sim \text{Beta}(a, b).$$

In the computations used for this project, the prior parameters were fixed at

$$a = b = 5,$$

so the prior is symmetric and concentrated around the middle of the interval rather than being strongly biased toward either endpoint. This is not meant to represent a deeply informed biological prior. Rather, it supplies a moderate starting point from which the BLAST evidence can update posterior support.

The likelihood model is intentionally simple and was chosen to match the final presentation and code used for the project. Conditional on π , the transformed evidence Y is modeled as

$$Y \mid \pi \sim \mathcal{N}(100\pi, \sigma^2),$$

with

$$\sigma^2 = 15.$$

This maps $\pi = 1$ to a mean transformed E-value of 100, corresponding to extremely strong BLAST evidence, while $\pi = 0$ maps to a mean of 0, corresponding roughly to E-values near 1. Larger values of π therefore correspond to stronger expected evidence. This is a deliberately simplified model. It is not presented as a complete latent-class model for homology. In particular, the project does not explicitly model a non-homolog class density f_0 in the denominator. Instead, it focuses on the numerator-style evidence term and treats the posterior only up to proportionality.

Under this choice, the unnormalized posterior density is

$$p(\pi \mid Y) \propto \pi^{a-1} (1 - \pi)^{b-1} \exp\left(-\frac{(Y - 100\pi)^2}{2\sigma^2}\right).$$

For a single observation this is the posterior used in the protein case. For the sequential fragment case, if the transformed evidence values are $Y_1, \dots, Y_k$, then the posterior becomes

$$p(\pi \mid Y_1, \dots, Y_k) \propto \pi^{a-1} (1 - \pi)^{b-1} \prod_{i=1}^k \exp\left(-\frac{(Y_i - 100\pi)^2}{2\sigma^2}\right).$$

This makes the sequential interpretation transparent: each new fragment multiplies the previous posterior by one more likelihood term.

Because the posterior is used only up to proportionality, Metropolis–Hastings is a natural computational tool. A random-walk proposal was used,

$$\pi^* = \pi^{(t)} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, s^2),$$

with proposal standard deviation

$$s = 0.05.$$

Proposals outside $(0, 1)$ were rejected automatically. Each chain used 20000 iterations with a burn-in of 3000. The project code then summarized the posterior using posterior means, standard deviations, and empirical 2.5% and 97.5% quantiles. The notebook also produced kernel-density plots of the posterior samples. The choice of Metropolis–Hastings was motivated by the fact that acceptance ratios depend only on posterior ratios, so the omitted normalizing constant cancels. [6, 7, 8]

5 Protein Case Study: Hemoglobin Subunit Beta

The hemoglobin protein example is the cleanest place to see how the model behaves in a fixed-target setting. Each query sequence was aligned against the original human beta-globin target, and the E-value of that target hit was recorded. The results are summarized in Table 1. As expected, the transformed posterior support declines as the mutation rate increases. The original human sequence and the 10% mutant both produce extremely small E-values

and posterior means near the top of the unit interval. By contrast, the 50% and 75% mutants fall much lower. The chicken beta-globin sequence remains strongly supported, which is biologically sensible: it is a real homolog, even though it is not identical to the human sequence.

Query	E-value	Mean	Std	$q_{0.025}$	$q_{0.975}$
Original human	4.0×10^{-107}	0.9438	0.0211	0.8958	0.9784
10% mutant	6.0×10^{-95}	0.8889	0.0307	0.8260	0.9460
25% mutant	6.0×10^{-72}	0.6996	0.0380	0.6254	0.7758
50% mutant	5.0×10^{-40}	0.3979	0.0386	0.3204	0.4744
75% mutant	7.0×10^{-8}	0.1205	0.0319	0.0600	0.1857
Chicken beta-globin	3.0×10^{-84}	0.8093	0.0351	0.7377	0.8775

Table 1: Posterior summaries for the hemoglobin beta fixed-target comparison. Posterior summaries were obtained from Metropolis–Hastings samples under the Beta–Normal model described in the methods section.

The main pattern is monotone but not trivial. The synthetic mutation ladder behaves as expected: as random substitutions accumulate, the E-value weakens and the posterior mean declines. However, the chicken sequence lies well above the heavily mutated synthetic sequences. This is one of the most useful outcomes of the case study, because it shows that the posterior summaries are not merely tracking the percentage of literal residue changes. Instead, the framework still recognizes a real homolog from another species as substantially more convincing than a severely corrupted artificial sequence.

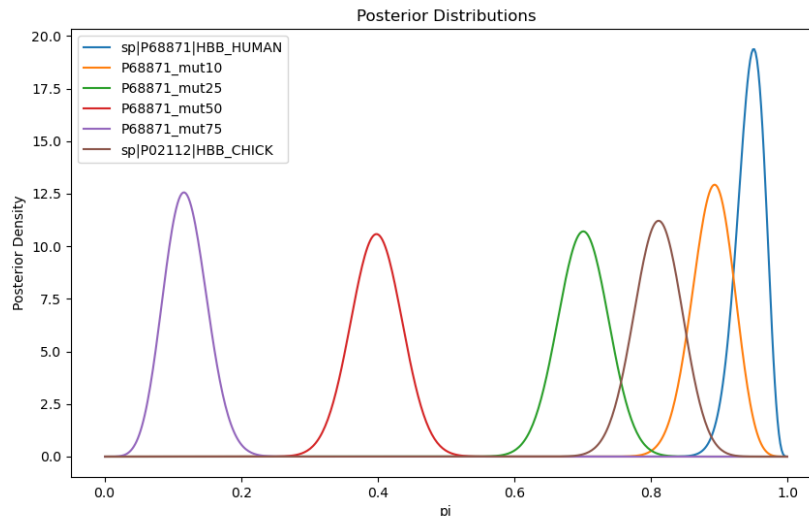


Figure 1: Posterior density curves for the hemoglobin beta protein case study. The original human sequence and the lightly mutated variants remain concentrated at high values of π , while heavier mutation shifts the posterior mass leftward. The chicken beta-globin sequence remains strongly supported despite being from a different species.

This protein example also shows one of the main interpretive advantages of the project. E-values by themselves already suggest a ranking, but the posterior summaries place all six queries on a common probabilistic scale. Means, standard deviations, and credible intervals are easier to compare directly than raw E-values spanning many orders of magnitude. At the same time, the example makes clear that the model is best viewed as a Bayesian reinterpretation of BLAST evidence, not as a fully calibrated biological truth model. The posterior numbers are only as meaningful as the structural assumptions built into the chosen prior and likelihood.

6 Sequential DNA Fragment Updating for Species Gene Matching

The second case study extends the same framework to a sequential updating problem. The full human beta-globin locus is long, so rather than using the entire 10106 bp sequence in one step, the analysis divides the locus into five 200 bp fragments. Each fragment is searched separately against species-filtered BLAST databases for human, chimpanzee, and mouse. The top-hit E-values for each species form a sequence of evidence contributions. Because the likelihood is multiplicative across observations, the posterior after fragment k becomes the prior-like starting point for fragment $k + 1$.

Table 2 reports both the fragment-level E-values and the final posterior summaries after all five fragments. Human receives consistently strong evidence at every step, with E-values of 7×10^{-100} across all five fragments. Its final posterior mean is therefore very high at 0.9596. Chimpanzee begins much more ambiguously, with the first fragment giving an E-value of 1, but later fragments become much stronger, eventually driving the posterior upward to a final mean of 0.7625. Mouse stays much weaker overall, despite two later fragments producing nontrivial hits, and ends with a much lower posterior mean of 0.2058.

Species	E_1	E_2	E_3	E_4	E_5	Mean	Std	$q_{0.025}$	$q_{0.975}$
Human	7×10^{-100}	7×10^{-100}	7×10^{-100}	7×10^{-100}	7×10^{-100}	0.9596	0.0125	0.9337	0.9823
Chimp	1	1×10^{-97}	4×10^{-88}	3×10^{-99}	7×10^{-101}	0.7625	0.0174	0.7287	0.7976
Mouse	1	1	1	5×10^{-60}	3×10^{-42}	0.2058	0.0169	0.1732	0.2395

Table 2: Sequential fragment E-values and final posterior summaries for the species gene-matching case study. Posterior summaries were computed after all five fragment updates.

The fragment-by-fragment posterior evolution is also informative visually. Figure 2 shows the posterior density progression for each species. The human posterior is already concentrated at high values after the first fragment and remains there. The chimpanzee posterior starts diffuse and much farther left, then shifts steadily right as strong later fragment hits accumulate. The mouse posterior moves only modestly and remains comparatively low. This is exactly the kind of sequential evidence behavior the Bayesian framework is designed to capture.

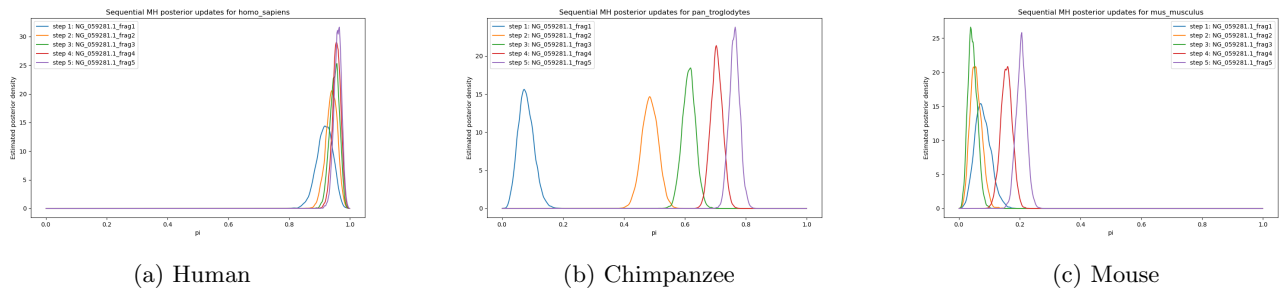


Figure 2: Sequential posterior evolution for species gene matching. Human evidence is consistently strong from the first fragment onward, chimpanzee evidence strengthens sharply after later fragments, and mouse remains much weaker overall.

This case study answers the main sequential question of the project. Rather than treating a long DNA sequence as one all-or-nothing object, the Bayesian formulation allows evidence to accumulate across fragments. This is especially appealing when some fragments are ambiguous and others are highly informative. The chimpanzee sequence illustrates this clearly: one weak fragment does not doom the analysis, because later strong fragments can still move the posterior substantially.

7 Conclusions, Limitations, and Future Work

The main conclusion of this project is that BLAST evidence can be re-expressed in a useful Bayesian language, even with a deliberately simple proof-of-concept model. In the hemoglobin protein case study, posterior summaries

track the expected decline in similarity across a mutation ladder while still assigning strong support to a real chicken homolog. In the DNA fragment case study, the same framework naturally supports sequential updating, with human, chimpanzee, and mouse showing three distinct evidence trajectories. The model therefore succeeds at the central interpretive goal of the project: it places BLAST-derived evidence on a posterior scale that is easier to summarize and compare than raw E-values alone.

At the same time, the current model has important limitations. The most serious is that it is not a full Bayesian generative model of homology. It does not explicitly model a non-homolog evidence distribution f_0 , and the mean structure 100π is an intentionally simple mapping rather than a biologically derived likelihood. The Beta prior parameters and the likelihood variance were fixed rather than estimated hierarchically. The protein case study also uses synthetic mutation ladders, which are helpful for controlled difficulty but do not mimic real evolutionary substitution processes. Finally, BLAST E-values are inherently database-dependent, so the numerical values reported here should be interpreted relative to the specific searches that generated them rather than as universal biological constants.

These limitations suggest several natural next steps. First, prior sensitivity analysis should be carried out to determine how much the posterior depends on the choice of a , b , and σ^2 . Second, the variance could be given its own prior so that posterior uncertainty in the spread of the likelihood is learned rather than fixed. Third, alternative likelihood forms could be explored, especially if transformed E-values show skewness or heavier tails than a normal model captures well. Fourth, translated search should be incorporated more seriously. In genomic DNA, issues such as frameshifts, exon–intron structure, and alternative splicing complicate direct comparison and would make a useful extension of the Bayesian framework. Finally, the most statistically ambitious next step would be to return to the fuller two-class formulation and explicitly model both homolog and non-homolog evidence rather than concentrating only on the homolog-like numerator term.

Despite its simplicity, the project demonstrates that Bayesian thinking can add interpretive value to standard BLAST output. BLAST already provides biologically meaningful similarity evidence. The contribution of the present work is to show how that evidence can be wrapped inside a posterior framework, summarized numerically, and extended naturally to sequential fragment updating.

References

- [1] Stephen F. Altschul et al. “Basic local alignment search tool”. In: *Journal of Molecular Biology* 215.3 (1990), pp. 403–410. DOI: 10.1016/S0022-2836(05)80360-2.
- [2] Stephen F. Altschul et al. “Gapped BLAST and PSI-BLAST: a new generation of protein database search programs”. In: *Nucleic Acids Research* 25.17 (1997), pp. 3389–3402. DOI: 10.1093/nar/25.17.3389.
- [3] The UniProt Consortium. “UniProt: the Universal Protein Knowledgebase in 2025”. In: *Nucleic Acids Research* 53.D1 (2025), pp. D609–D617. DOI: 10.1093/nar/gkae1010.
- [4] Nuala A. O’Leary et al. “Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation”. In: *Nucleic Acids Research* 44.D1 (2016), pp. D733–D745. DOI: 10.1093/nar/gkv1189.
- [5] Steven Henikoff and Jorja G. Henikoff. “Amino acid substitution matrices from protein blocks”. In: *Proceedings of the National Academy of Sciences* 89.22 (1992), pp. 10915–10919. DOI: 10.1073/pnas.89.22.10915.
- [6] Nicholas Metropolis et al. “Equation of state calculations by fast computing machines”. In: *The Journal of Chemical Physics* 21.6 (1953), pp. 1087–1092. DOI: 10.1063/1.1699114.
- [7] W. K. Hastings. “Monte Carlo sampling methods using Markov chains and their applications”. In: *Biometrika* 57.1 (1970), pp. 97–109. DOI: 10.1093/biomet/57.1.97.
- [8] Andrew Gelman et al. *Bayesian Data Analysis*. 3rd ed. Boca Raton, FL: CRC Press, 2013.